

GainOne AI Customer Support System
Software Requirements Specification
(SRS, HLD, Development & Testing)



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Level 1 – Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the basic software requirements for the GainOne AI Customer Support System. The system aims to automate customer support by providing quick, accurate, and intelligent responses using Artificial Intelligence. This document serves as the foundation for understanding the project before moving to design and development.

1.2 Scope

The GainOne AI Customer Support System is designed to provide 24/7 automated customer support through an AI-powered chatbot. The system helps customers resolve common queries, manages support tickets, maintains a knowledge base, and provides an administrative dashboard for monitoring customer interactions.

1.3 Overview

The system combines Artificial Intelligence with customer support automation. It allows customers to interact with an AI chatbot, receive instant responses, and escalate complex issues to human support agents when required.

2. General Description

2.1 Product Perspective

The GainOne AI Customer Support System is an intelligent support platform developed for organizations to improve customer service. It reduces manual effort, improves response time, and enhances customer satisfaction.

2.2 Product Functions

- AI Chatbot
- Customer Support
- Ticket Management
- Knowledge Base
- Customer Authentication
- Admin Dashboard
- Reports and Analytics

2.3 User Characteristics

- Customer
- Support Agent
- Administrator

2.4 Assumptions

- Users have internet connectivity.
- Users possess valid login credentials.
- The knowledge base contains updated information.
- AI services are available continuously.

3. Functional Requirements

The system shall:

1. Accept customer queries.
2. Generate AI-based responses.
3. Search information from the knowledge base.
4. Create support tickets when required.
5. Escalate conversations to human agents.
6. Allow administrators to manage customer information.
7. Generate reports.
8. Maintain conversation history.
9. Generate a summary of each client conversation and automatically send/store it in the company's official email for record-keeping and future reference.

4. User Interface Requirements

The software provides:

- Customer Login Page
- AI Chat Interface
- Customer Dashboard
- Admin Dashboard
- Ticket Management Screen
- Knowledge Base Management Screen
- Reports Screen

The interface should be simple, responsive, and easy to use.

5. Performance Requirements

The system should:

- Respond within a few seconds.
- Support multiple users simultaneously.
- Provide continuous availability.
- Handle increasing customer requests efficiently.

6. Non-Functional Requirements

Usability

- Easy-to-use interface.
- User-friendly navigation.

Reliability

- High system availability.
- Consistent response generation.

Security

- Secure authentication.
- Role-based access.
- Data encryption.
- Secure communication.

7. Schedule

Project Timeline

Phase	Duration
Requirement Analysis	1 Week
SRS Preparation	1 Week
Review & Approval	1 Week

8. Appendices

Glossary

Term	Description
AI	Artificial Intelligence
Chatbot	Automated conversational assistant
Knowledge Base	Collection of documents used for answering queries

Term	Description
Ticket	Customer support request
Administrator	System manager
Support Agent	Human customer support executive

Level 2 – High-Level Design (HLD) Document

Introduction

1. Purpose

The purpose of the High-Level Design (HLD) document is to provide a clear overview of the architecture and major components of the GainOne AI Customer Support System. It explains how different modules interact with each other and how customer requests are processed. This document acts as a bridge between the Software Requirements Specification (SRS) and the detailed system implementation.

2. Scope

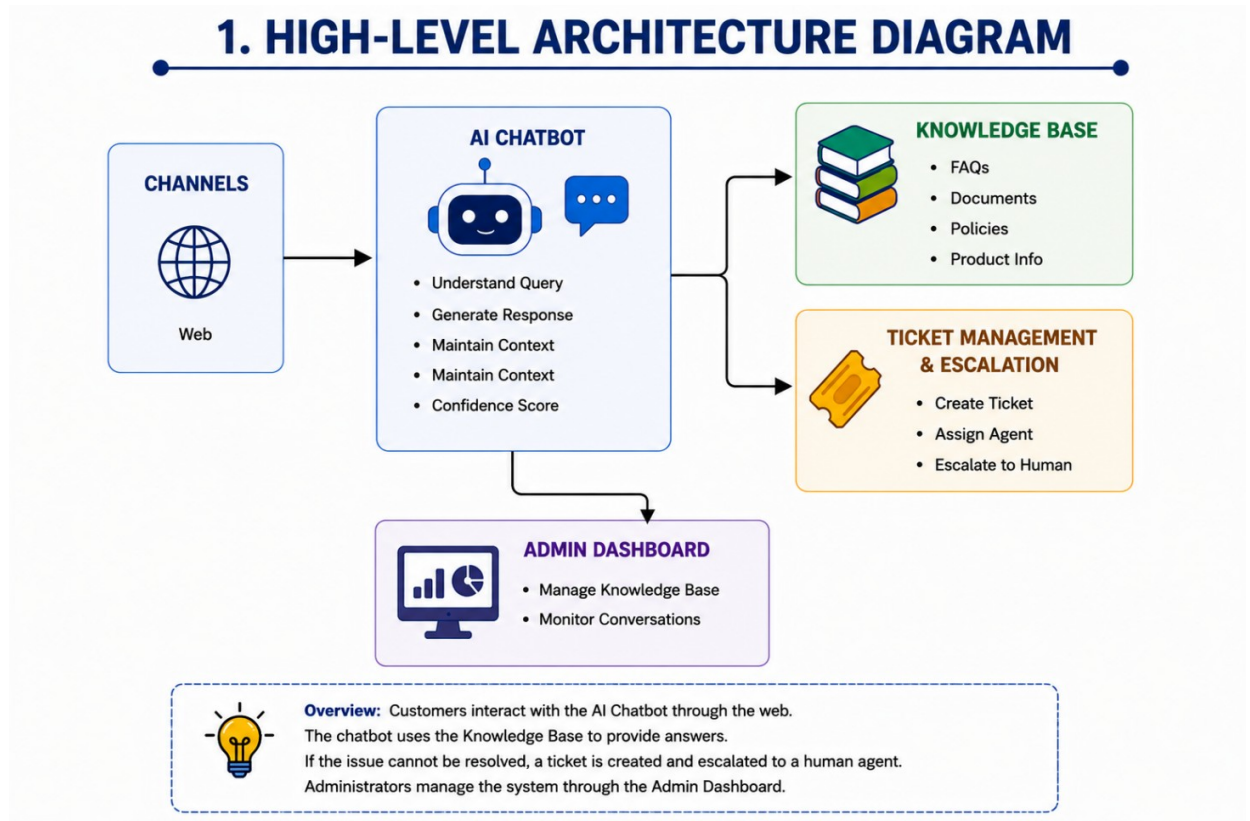
This document covers the overall architecture of the GainOne AI Customer Support System. It includes the system architecture, block diagram, workflow, context diagram, and module diagram. It describes the major software components and their interactions without discussing programming code or database implementation.

3. Design Objectives

The High-Level Design aims to:

- Provide a clear understanding of the system architecture.
- Describe the interaction between different modules.
- Explain the flow of customer requests.
- Identify the major software components.
- Support future development and implementation.

1.High-Level Architecture Diagram



Description

The High-Level Architecture Diagram provides an overview of the GainOne AI Customer Support System. It illustrates the major software components and how they interact to process customer requests. The system is designed to provide fast, intelligent, and efficient customer support using Artificial Intelligence.

The architecture begins with the customer accessing the system through the web application. The customer submits a query, which is received and processed by the AI Chatbot. The chatbot understands the user's request and searches for relevant information from the Knowledge Base.

The Knowledge Base stores frequently asked questions (FAQs), company documents, policies, and product information. The chatbot retrieves the most relevant information and generates an appropriate response for the customer.

If the chatbot cannot provide a satisfactory answer or the confidence score is low, the request is forwarded to the Ticket Management & Escalation module. This module

creates a support ticket, assigns it to a support agent, and escalates the issue for manual resolution.

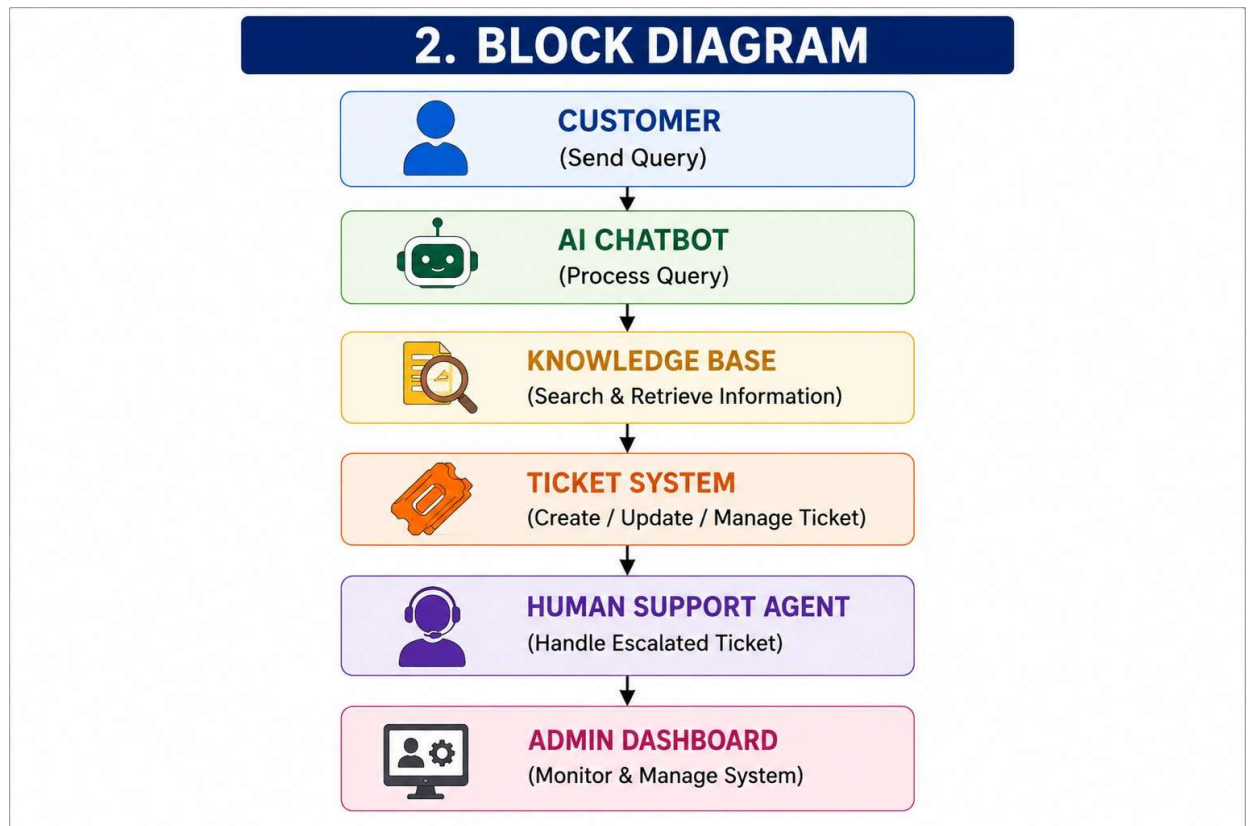
The Admin Dashboard allows administrators to manage users, maintain the knowledge base, monitor customer conversations, and configure system settings. It serves as the central control panel for managing and monitoring the entire application.

Overall, the architecture ensures that customer queries are handled efficiently while providing seamless integration between the chatbot, knowledge base, ticket management, and administration modules.

Main Components

- Customer Channels
- AI Chatbot
- Knowledge Base
- Ticket Management
- Admin Dashboard

1. Block Diagram



Description

The Block Diagram illustrates the overall operational flow of the GainOne AI Customer Support System. It shows how customer requests are processed through different modules, starting from the customer and ending with the administrator managing the system.

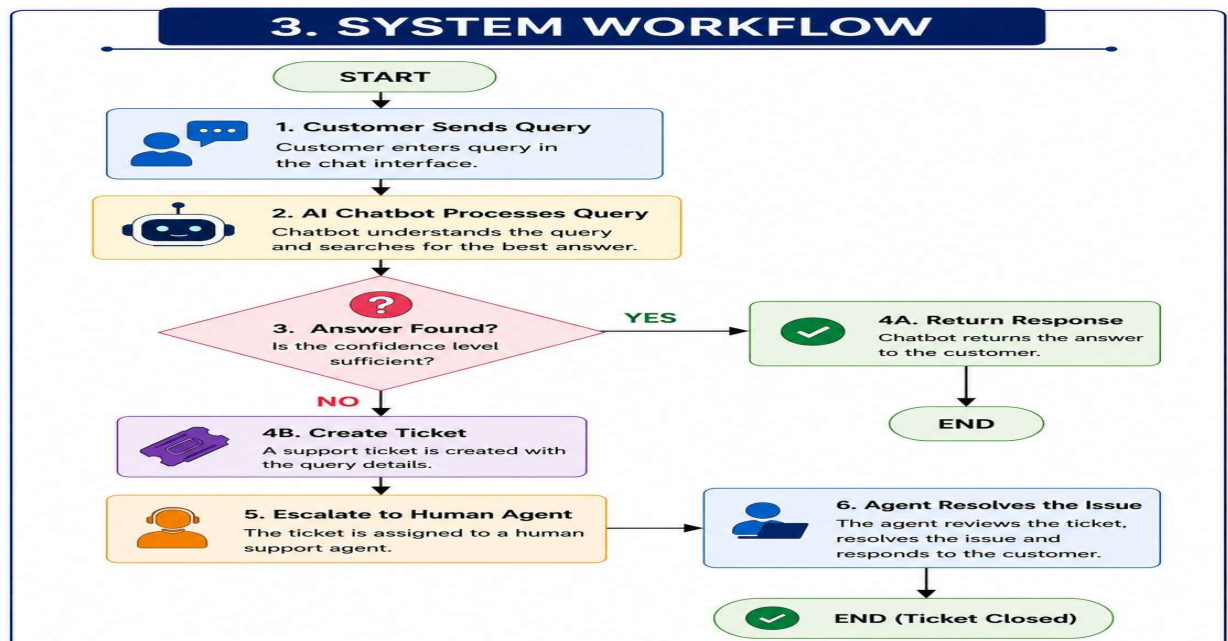
Initially, the Customer submits a query through the web interface. The request is received by the AI Chatbot, which processes the query using Artificial Intelligence techniques. The chatbot then searches the Knowledge Base to retrieve relevant information such as FAQs, company documents, and product details.

If the chatbot successfully finds an appropriate answer, it responds directly to the customer. If the chatbot cannot answer the query with sufficient confidence, the request is forwarded to the Ticket System. The Ticket System creates a support ticket and manages its status throughout the resolution process.

The ticket is then assigned to a Human Support Agent, who reviews the issue, communicates with the customer if necessary, and provides the appropriate solution. After resolving the issue, the support agent updates the ticket status.

Finally, the Admin Dashboard enables administrators to monitor the entire system, manage users, update the knowledge base, supervise customer interactions, and configure system settings. This ensures smooth system operation and efficient customer support.

2. System Workflow



Description

The System Workflow explains how customer requests are processed step by step.

Initially, the customer submits a query through the chatbot interface. The chatbot processes the query using Artificial Intelligence and searches the knowledge base.

If the chatbot finds a suitable answer with sufficient confidence, it immediately returns the response to the customer.

If the chatbot cannot answer the question, it automatically creates a support ticket and forwards it to a human support agent.

The support agent reviews the issue, communicates with the customer if necessary, resolves the problem, and closes the support ticket.

Workflow Steps

- 10. Customer sends a query.
- 11. AI Chatbot processes the query.
- 12. Knowledge Base is searched.
- 13. Decision is made based on confidence score.
- 14. Return AI response or create support ticket.
- 15. Human agent resolves the issue.
- 16. Ticket is closed.

4.Technology Stack



The GainOne AI Customer Support System is developed using modern technologies to provide a fast, scalable, and intelligent customer support platform. It leverages **Retrieval-Augmented Generation (RAG)** to deliver accurate and context-aware responses.

LAYER	TECHNOLOGY	ICON	PURPOSE
 Frontend	React.js	 React	Develops the web-based user interface.
 Backend	FastAPI	 FastAPI	Handles API requests and business logic.
 Programming Language	Python	 python	Implements AI logic and backend services.
 AI Framework	LangChain	 LangChain	Manages AI workflows and integrates LLM with external data sources.
 Retrieval-Augmented Generation (RAG)	RAG Pipeline	 RAG	Combines retrieved information from the knowledge base with LLM to generate accurate and context-aware responses.
 Large Language Model	Groq API / Llama 3.1	 groq Llama 3.1	Generates intelligent responses to customer queries.
 Vector Database	FAISS	 FAISS	Stores and retrieves document embeddings efficiently for semantic search.
 Database	MySQL	 MySQL	Stores user accounts, tickets, and system data.
 Authentication	JWT	 JWT	Provides secure user authentication and authorization.
 Document Processing	PyPDF2 / python-docx		Extracts text from PDF and Word documents.
 Embedding Model	HuggingFace Sentence Transformers	 HUGGING FACE	Converts text into vector embeddings for semantic search.
 Version Control	Git & GitHub		Tracks source code changes and enables collaboration.
 Development Tools	Visual Studio Code	 VS Code	Used for coding, testing, and debugging the application.



Note: The system follows a RAG (Retrieval-Augmented Generation) approach where the AI retrieves relevant information from the knowledge base (using FAISS) and uses it along with the LLM (via Groq API / Llama 3.1) to generate accurate and reliable answers.

RAG FLOW OVERVIEW



Description

The Technology Stack represents the collection of software technologies, frameworks, programming languages, databases, and tools used to develop the GainOne AI Customer Support System. These technologies work together to provide an intelligent, secure, and scalable customer support platform.

The system follows a Retrieval-Augmented Generation (RAG) architecture, where the AI chatbot retrieves relevant information from the Knowledge Base before generating a response. This approach improves the accuracy of responses by combining document retrieval with the reasoning capabilities of a Large Language Model (LLM).

The frontend of the application is developed using React.js, which provides a responsive and user-friendly web interface. The backend is implemented using FastAPI, which manages API requests, business logic, and communication between different system components.

The core application logic is written in Python, while LangChain is used to integrate the chatbot with the RAG pipeline and manage AI workflows.

The RAG Pipeline retrieves relevant information from the Knowledge Base using FAISS, a vector database optimized for semantic similarity searches. The retrieved content is then passed to the Groq API (Llama 3.1), which generates accurate and context-aware responses for customer queries.

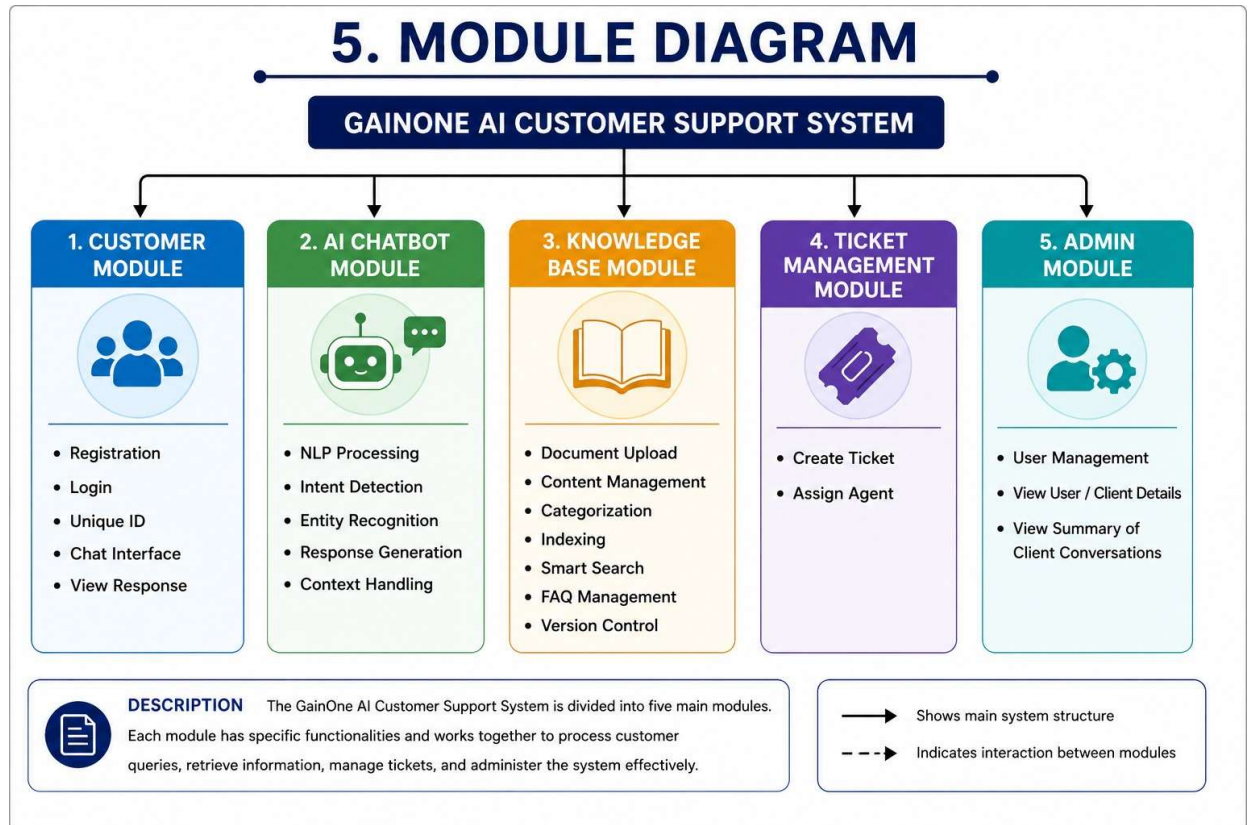
Customer details, support tickets, and system information are stored in a MySQL database. User authentication is secured using JSON Web Token (JWT) to ensure authorized access.

Documents uploaded to the system are processed using PyPDF2 and python-docx, allowing text extraction from PDF and Word files. The extracted text is converted into vector embeddings using HuggingFace Sentence Transformers, enabling efficient semantic search within the knowledge base.

The project is managed using Git and GitHub for version control, while Visual Studio Code is used as the primary development environment.

Overall, this technology stack provides a robust foundation for developing an AI-powered customer support system that is scalable, efficient, secure, and capable of delivering accurate responses through Retrieval-Augmented Generation (RAG).

5. Module Diagram



Description

The Module Diagram represents the overall structure of the GainOne AI Customer Support System by dividing it into five major functional modules. Each module performs a specific responsibility and works together with the other modules to provide an efficient AI-based customer support solution. The modular design improves system organization, simplifies maintenance, and allows future enhancements without affecting the entire application.

The Customer Module serves as the entry point of the system, where customers can register, log in, access the chat interface, and submit their queries. Every registered customer is assigned a unique user ID, which helps maintain individual chat history and user information.

The AI Chatbot Module is the core processing component of the system. It uses Natural Language Processing (NLP) to understand customer queries, detect user intent, recognize important entities, maintain conversation context, and generate intelligent responses using the RAG-based AI model.

The Knowledge Base Module acts as the central repository of information. It stores company documents, FAQs, policies, and product-related information. Documents are uploaded, categorized, indexed, and managed so that the AI chatbot can quickly retrieve relevant information during customer conversations.

The Ticket Management Module handles customer requests that cannot be resolved automatically by the AI chatbot. It creates support tickets for unresolved issues and assigns them to the appropriate support agent for manual resolution. This ensures that customer problems are addressed efficiently and tracked until completion.

The Admin Module provides complete administrative control over the system. Administrators can manage users, monitor customer details, review conversation summaries, and oversee system activities. This module helps maintain the knowledge base and ensures smooth operation of the customer support platform.

Together, these five modules work in an integrated manner to automate customer support, reduce manual effort, improve response accuracy, and provide a better customer experience.

6. Advantages of High-Level Design

The High-Level Design offers several advantages:

- Provides a clear understanding of the complete system.
- Improves communication among developers and stakeholders.
- Identifies major modules before development.
- Simplifies future maintenance and upgrades.
- Reduces development complexity.
- Helps developers understand the interaction between system components.

7. Conclusion

The High-Level Design document presents the overall architecture and design of the GainOne AI Customer Support System. It explains the major software modules, system workflow, and interactions between users and system components. This document serves as the foundation for the next phase, which focuses on Low-Level Design (LLD), database design, and implementation.

Level 3 – Development Document

1. Introduction

The Development phase focuses on implementing the GainOne AI Customer Support System according to the Software Requirements Specification (SRS) and High-Level Design (HLD). The system was developed using modern web technologies and Artificial Intelligence to automate customer support while providing an efficient administrative interface.

2. Development Environment

Component	Technology
Programming Language	Python
Backend Framework	FastAPI
Frontend	HTML, CSS, JavaScript
Database	MySQL
AI Framework	LangChain
LLM	Groq (Llama 3.1)
Vector Database	FAISS
Authentication	JWT
IDE	Visual Studio Code
Version Control	Git & GitHub

3. Project Structure

```
GainOne_AI_Customer_Support_System

backend/
  app/
  authentication/
  chatbot/
  customer/
  admin/
  routes/
  services/
  rag/
  tickets/
```

```
uploads/  
vector_db/  
  
frontend/  
  login.html  
  register.html  
  dashboard.html  
  profile.html  
  chat.html  
  tickets.html  
  admin_login.html  
  admin_dashboard.html
```

4. Modules Developed

Customer Module

Features:

- Customer Registration
- Customer Login
- Dashboard
- Profile
- AI Chat
- Ticket History
- Logout

AI Chatbot Module

Features:

- Natural Language Processing
- RAG Search
- FAISS Retrieval
- Groq Response Generation
- Conversation Storage
- Chat History

Knowledge Base Module

Features:

- PDF Upload
- Word Document Upload

- Text Extraction
- Vector Embedding Generation
- Semantic Search

Ticket Management Module

Features:

- Automatic Ticket Creation
- Human Escalation
- Ticket Assignment
- Ticket Status Update
- Ticket Closing

Admin Module

Features:

- Admin Login
- Dashboard
- Customer Management
- View Customer Conversations
- Ticket Management
- Monitor AI Chat

5. Database Tables

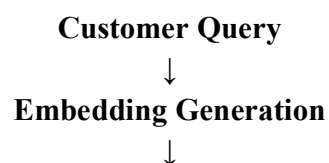
The project uses MySQL with the following tables:

- users
- admin
- tickets
- chat_history
- knowledge_base

6. AI Development

The chatbot follows Retrieval-Augmented Generation (RAG).

Workflow





7. Security

Implemented security features:

- JWT Authentication
- Password Hashing (bcrypt)
- Role-based Login
- Protected APIs
- Secure Database Access

8. Additional Features

Implemented features:

- Human Support Escalation
- Automatic Ticket Creation
- AI Conversation Summary
- Email Notification
- Customer Chat History
- Admin Dashboard Analytics

9. Development Outcome

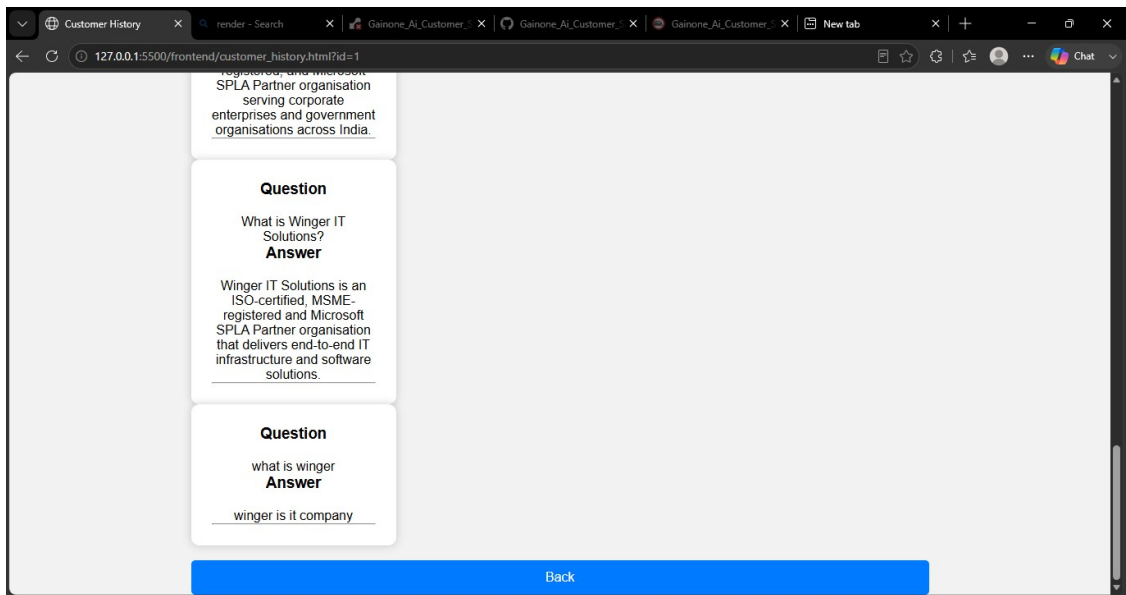
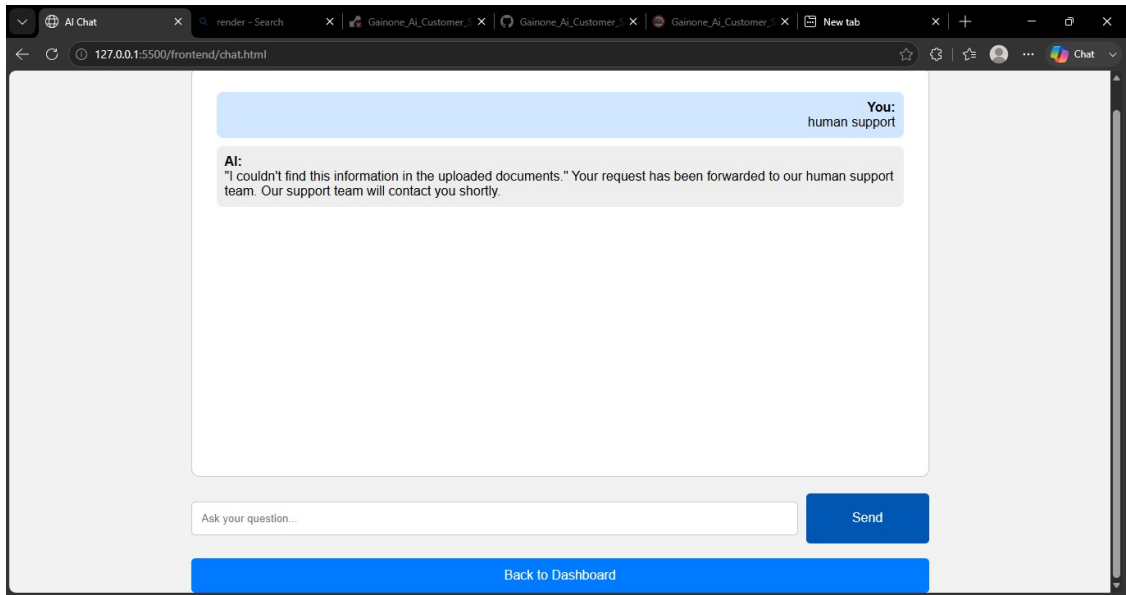
Successfully developed:

- AI Customer Portal
- Admin Portal
- RAG Chatbot
- Ticket Management System
- Email Summary System
- Human Escalation System

Output Screenshot:

A screenshot of a web browser displaying the login page for GainOne AI Customer Support. The browser's address bar shows the URL `127.0.0.1:5500/frontend/login.html`. The page features a central white card with a blue header "GainOne AI Customer Support". Below the header are two input fields: "Enter Email" and "Enter Password". A blue "Login" button is positioned below the password field. At the bottom of the card, there is a link that says "Don't have an account? [Register](#)".

A screenshot of a web browser displaying the registration page for GainOne AI Customer Support. The browser's address bar shows the URL `127.0.0.1:5500/frontend/register.html`. The page features a central white card with a blue header "Create Account". Below the header are three input fields: "Enter Name", "Enter Email", and "Enter Password". A blue "Register" button is positioned below the password field. At the bottom of the card, there is a link that says "Already have an account? [Login](#)".



Admin - All Tickets

render - Search

Gainone_AI_Customer...

Gainone_AI_Customer...

Gainone_AI_Customer...

New tab

127.0.0.1:5500/frontend/tickets_admin.html

Chat

All Customer Tickets

← Back to Dashboard

Logout

Ticket ID	Customer	Email	Issue	Status	Priority	Assigned Agent	
1	Lakshman	lakshman@gmail.com	Chatbot could not answer my question	high	Open	Not Assigned	Close
2	keerti	keerti@gmail.com	Welcome to GainOne AI Customer Support	Low	Open	Nandita	Close
3	faisal	faisal@gmail.com	Welcome to GainOne AI Customer Support	Low	Open	Nandita	Close

GainOne Admin Dashb...

render - Search

Gainone_AI_Customer...

Gainone_AI_Customer...

Gainone_AI_Customer...

New tab

127.0.0.1:5500/frontend/admin_dashboard.html

Chat

Welcome, Nandita

Logout

Total Customers

5

Total Chats

21

Total Tickets

3

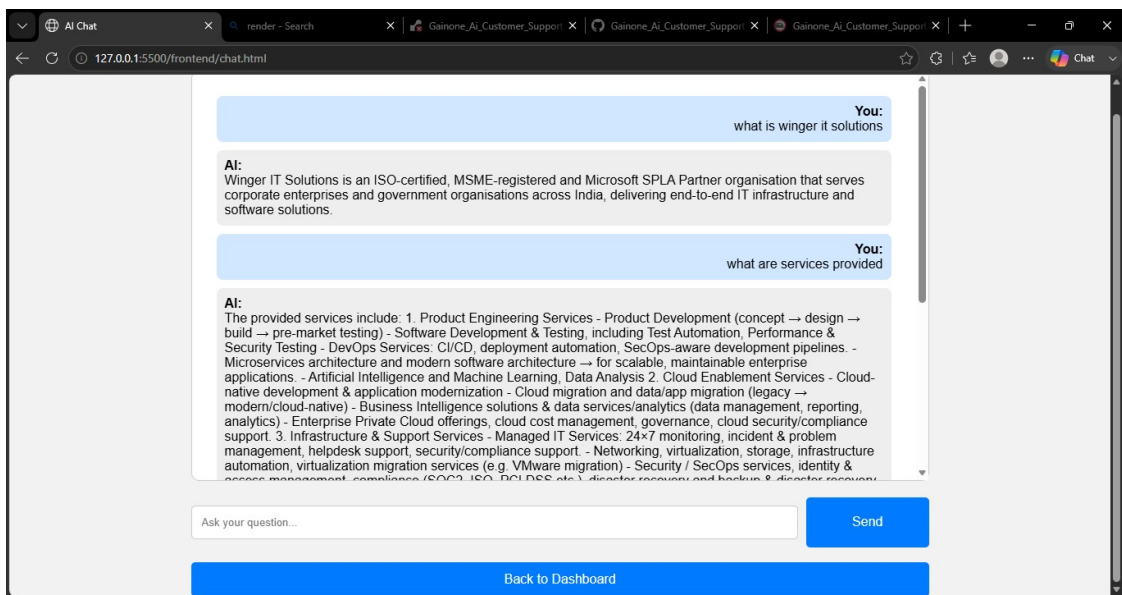
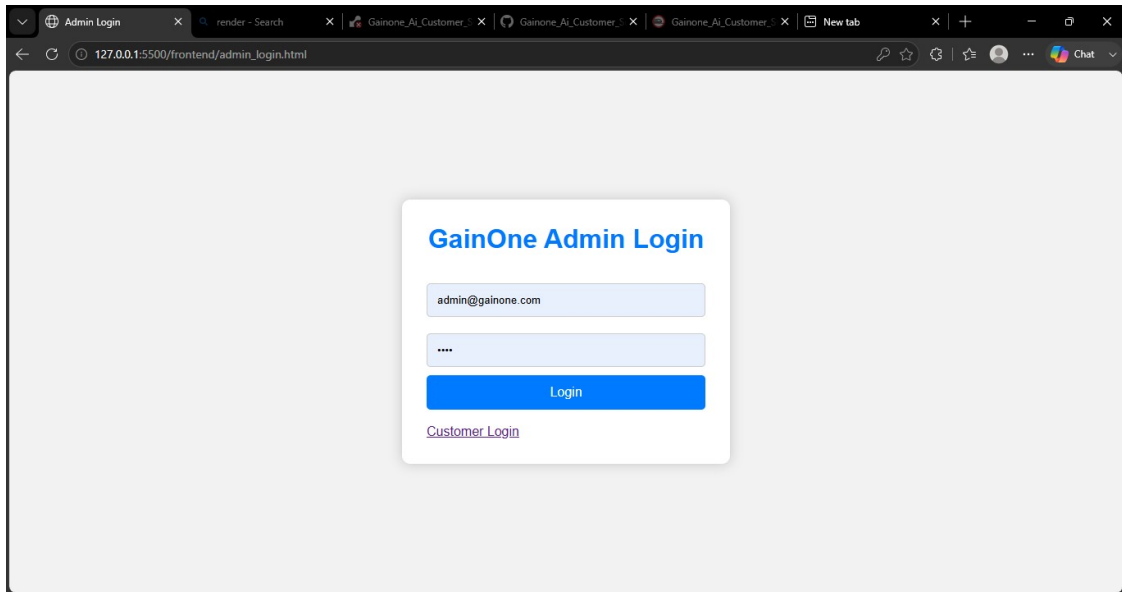
Total Documents

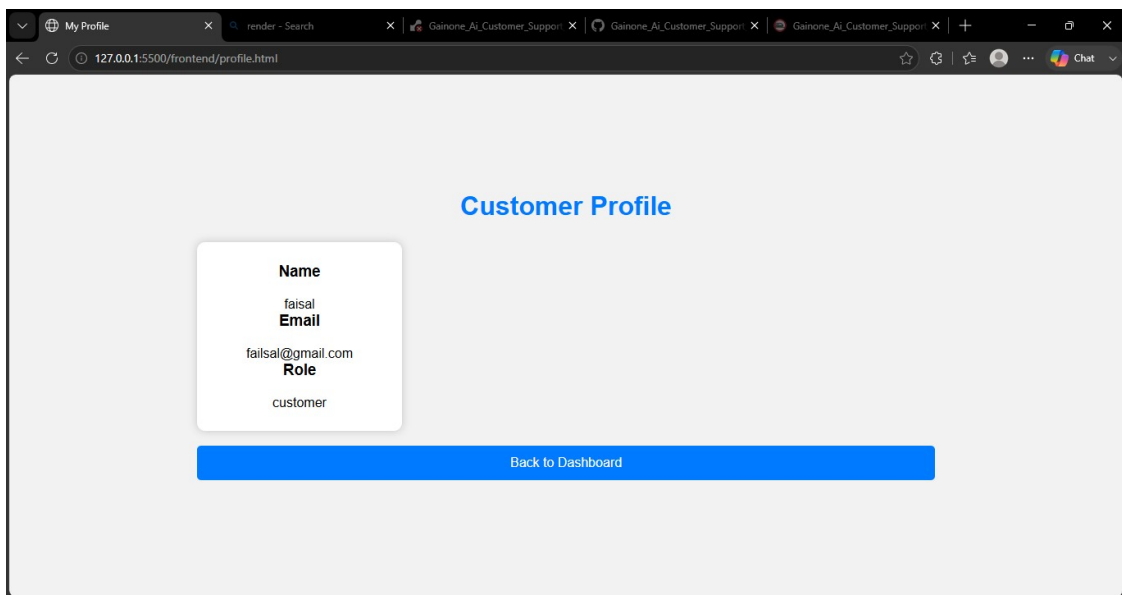
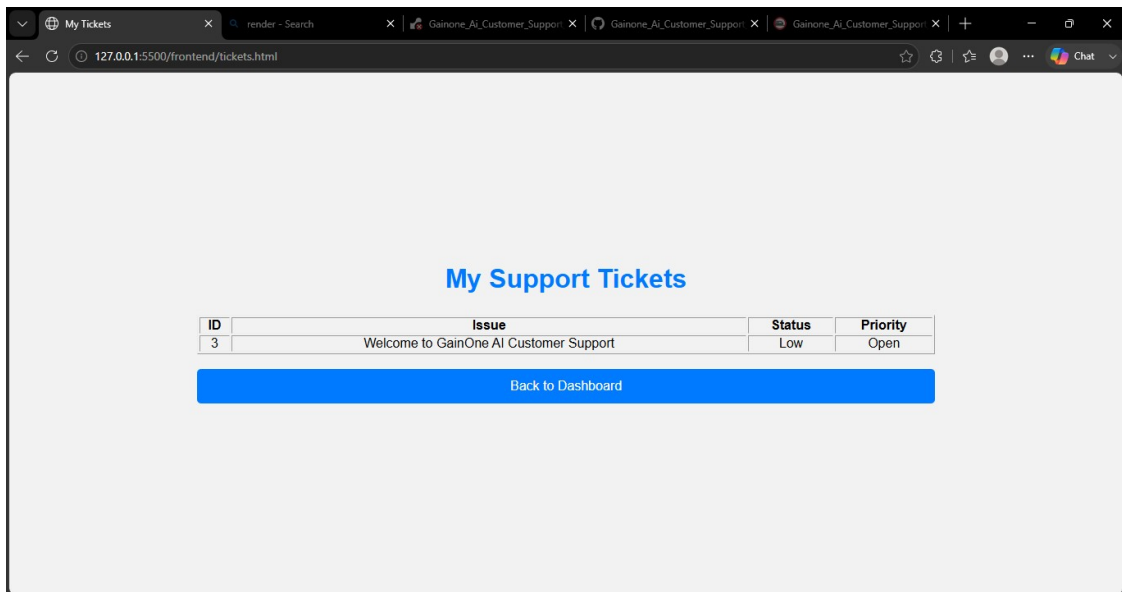
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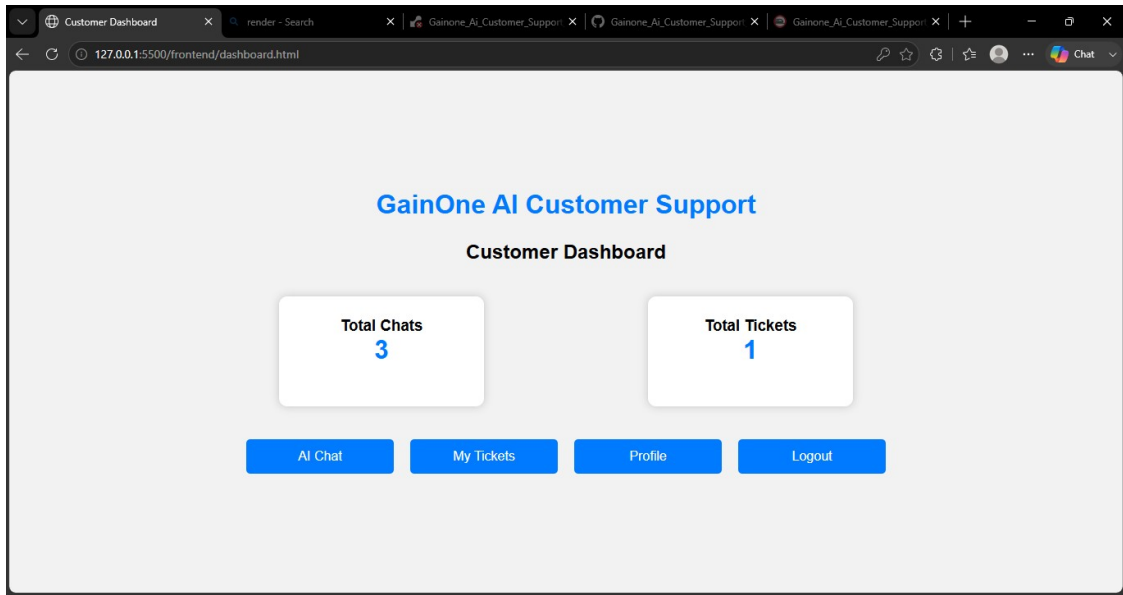
Registered Customers

ID	Name	Email	Total Chats	Total Tickets	History
1	Lakshman	lakshman@gmail.com	12	1	View
2	nandu	nandu@gmail.com	0	0	View
3	vamshith	vamshith@gmail.com	1	0	View
4	keerti	keerti@gmail.com	3	1	View
5	faisal	faisal@gmail.com	5	1	View

View All Tickets







Level 4 – Testing Document

1. Introduction

The testing phase verifies that every module functions according to the project requirements. Functional testing was performed on customer features, admin features, AI chatbot responses, authentication, and ticket management.

2. Testing Environment

Component	Details
Browser	Microsoft Edge
Backend	FastAPI
Database	MySQL
API Testing	Browser + FastAPI
Operating System	Windows 11

3. Functional Test Cases

Test Case	Expected Result	Status
Customer Registration	User Registered Successfully	Pass
Duplicate Registration	Email Already Exists	Pass
Customer Login	Login Successful	Pass
Invalid Login	Invalid Password	Pass
Customer Dashboard	Dashboard Loaded	Pass
View Profile	Customer Details Displayed	Pass
AI Chat	AI Generated Response	Pass
Human Support Request	Ticket Created Automatically	Pass
View Tickets	Ticket Displayed	Pass
Logout	User Logged Out Successfully	Pass
Admin Login	Login Successful	Pass
Admin Dashboard	Dashboard Loaded	Pass
View Customers	Customer List Displayed	Pass

Test Case	Expected Result	Status
Customer History	Chat History Displayed	Pass
Ticket Management	Ticket Closed Successfully	Pass

4. AI Testing

Test	Result
Known Question	AI Response Generated
Unknown Question	Human Escalation Triggered
Document Retrieval	Successful
FAISS Search	Successful
Groq Response	Successful

5. Database Testing

Verified:

- User Data Stored
- Chat History Stored
- Ticket Information Stored
- Admin Details Retrieved
- Knowledge Base Retrieved

Status: Passed

6. Security Testing

Performed:

- Login Authentication
- JWT Validation
- Password Hashing
- Unauthorized Access Prevention

Status: Passed

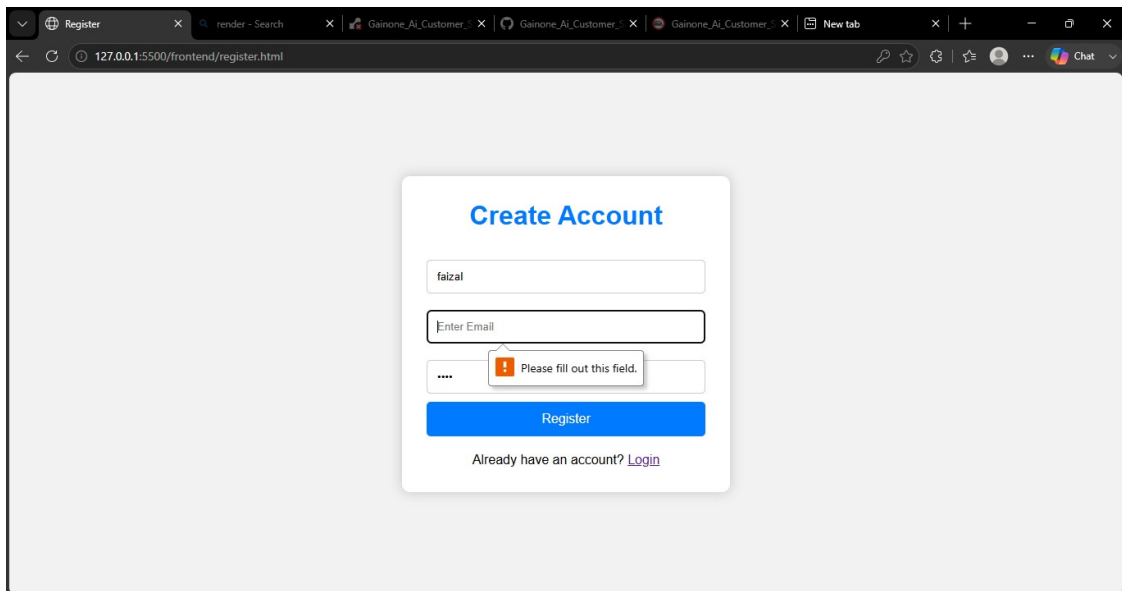
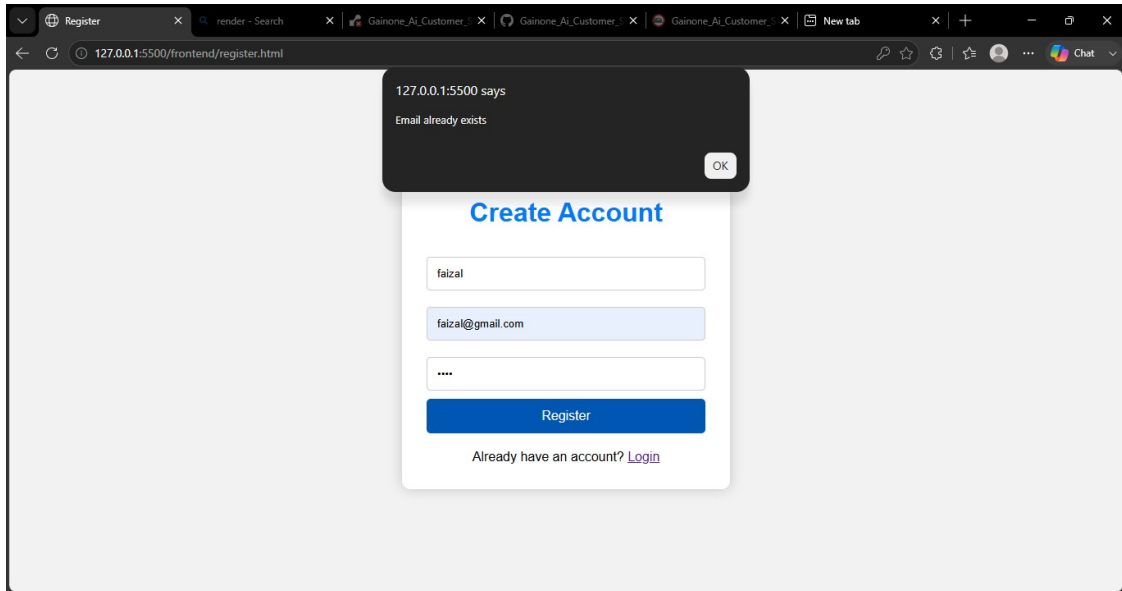
7. Performance Testing

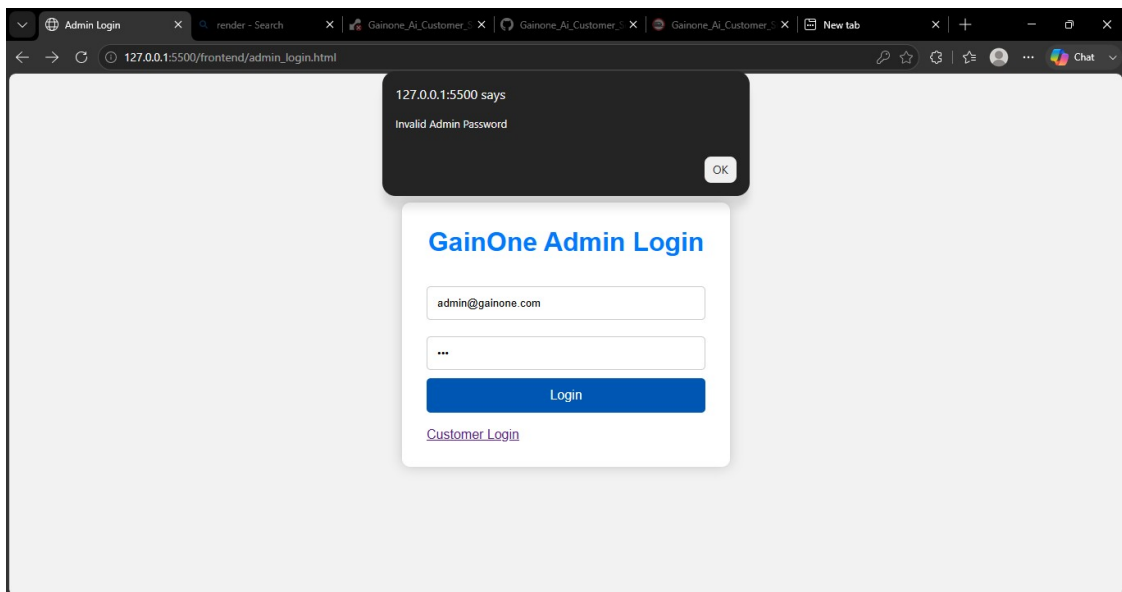
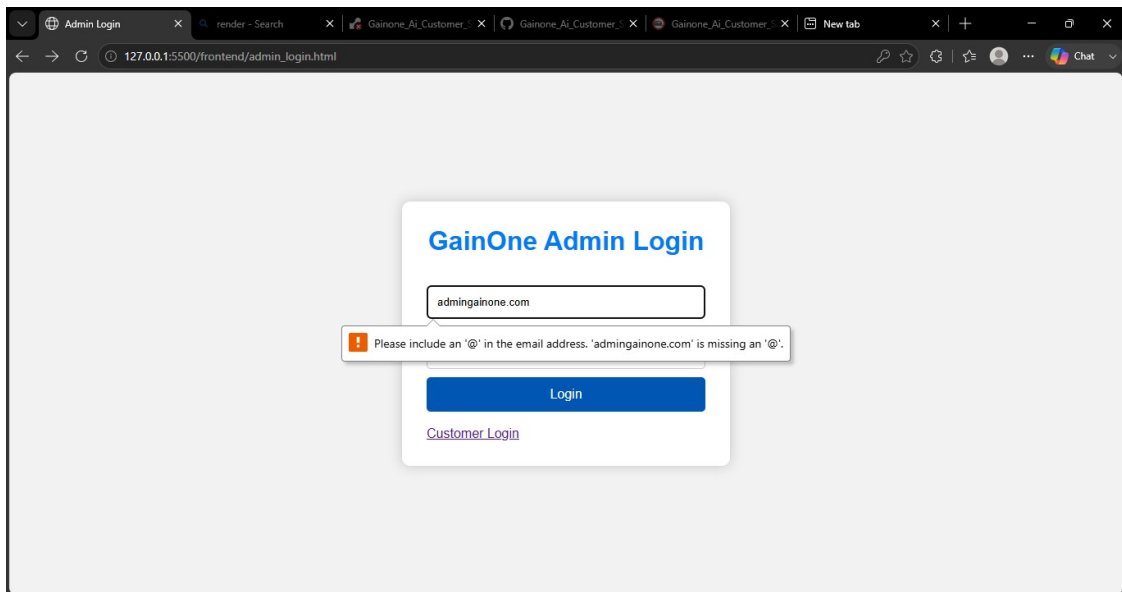
Observed:

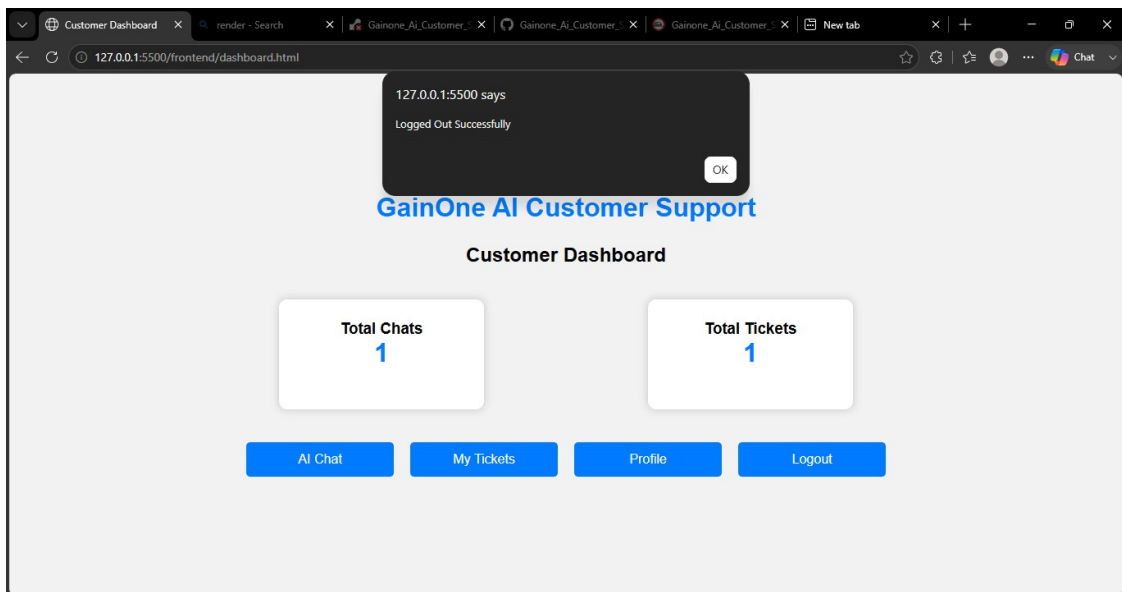
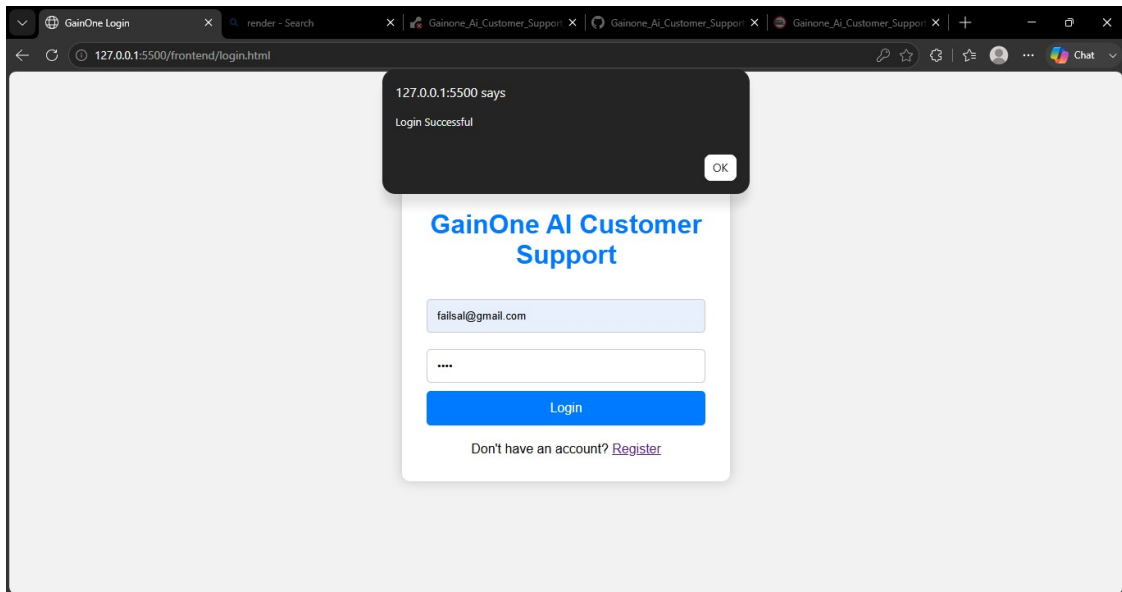
- Login Response Time: Less than 2 seconds
- AI Response: 2–5 seconds

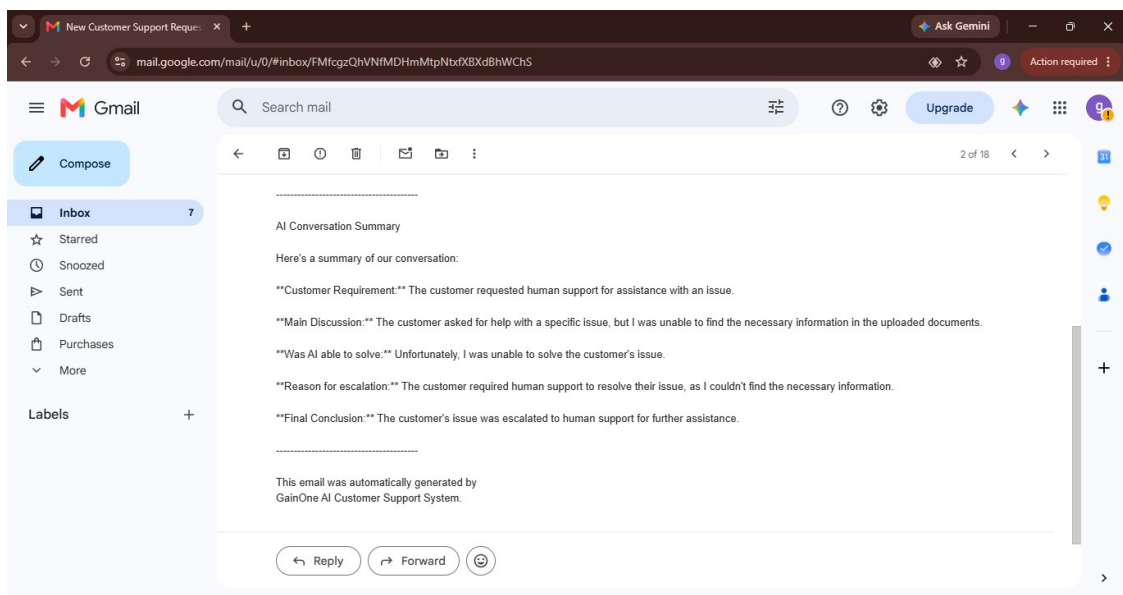
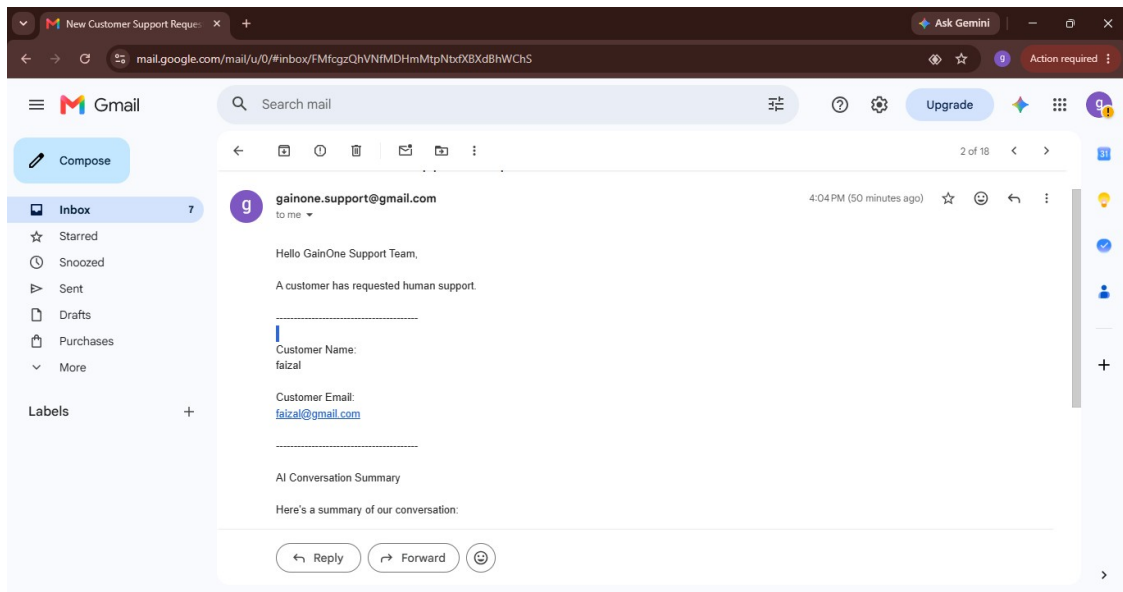
- Dashboard Loading: Less than 3 seconds

Output Screenshot:









Conclusion:

The GainOne AI Customer Support System was successfully designed, developed, and tested as an intelligent web-based customer support platform. The project combines Artificial Intelligence, Retrieval-Augmented Generation (RAG), and Large Language Models (Groq Llama 3.1) to provide fast, accurate, and automated responses to customer queries. By integrating a Knowledge Base, FAISS vector database, and LangChain, the chatbot retrieves relevant information from uploaded documents before generating responses, improving the quality and reliability of customer support.

The system includes complete customer and administrator modules with secure authentication, AI-powered chat, ticket management, conversation history, profile management, and an administrative dashboard. When the AI chatbot cannot resolve a customer's query, the system automatically creates a support ticket, generates a conversation summary, and escalates the issue to the human support team through email, ensuring uninterrupted customer service.

The application was developed using FastAPI, Python, HTML, CSS, JavaScript, MySQL, JWT Authentication, and GitHub for version control. Functional, security, database, and user interface testing confirmed that all modules operate as expected. The project successfully demonstrates the practical application of Artificial Intelligence in customer support automation while reducing manual effort, improving response time, and enhancing customer satisfaction.

Overall, the GainOne AI Customer Support System provides a scalable, secure, and efficient solution for modern customer service and serves as a strong foundation for future enhancements such as cloud deployment, multilingual support, voice-based interaction, analytics dashboards, and integration with CRM and messaging platforms.

GitHub Repository

Project Repository:

https://github.com/Lakshman0707/Gainone_Ai_Customer_Support_System.git